Lines and Shapes

Lines, rays, parallel and perpendicular, and lines of symmetry.

Segment, ray, line



line segment: 2 endpoints



ray: 1 endpoint

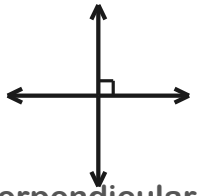


line: no endpoints

Parallel and perpendicular

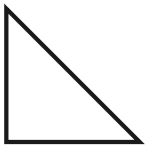


Parallel: never meet.



Perpendicular: a square corner.

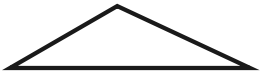
Triangles by their biggest angle



right

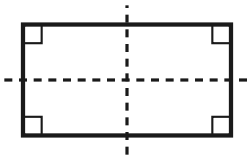


acute



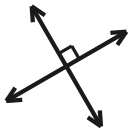
obtuse

Lines of symmetry: fold it and the halves match



A rectangle has 2 lines of symmetry.

Your turn!



parallel (perpendicular) neither

For teachers

Goal: points, lines, segments, rays, angles, parallel and perpendicular lines, triangles by angle, and lines of symmetry (4.G.1–3). Use a square corner (a sticky note) to test angles.

Lines and Shapes: Practice

Look for endpoints, square corners and matching halves.

Part 1 Segment, ray or line? Count the endpoints.

1)  segment **(ray)** line

2) **(segment)** ray line

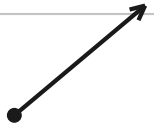
3)  segment ray **(line)**

4)  **(segment)** ray line

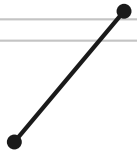
5)  segment ray **(line)**

6)  segment **(ray)** line

7)  segment ray **(line)**

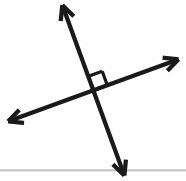
8)  segment **(ray)** line

9)  **(segment)** ray line

10)  **(segment)** ray line

Lines and Shapes: Practice (continued)

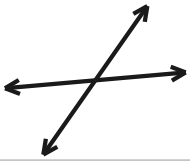
Part 2 Parallel, perpendicular or neither? Circle one.

11) 

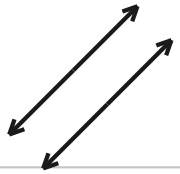
parallel **(perpendicular)** neither

12) 

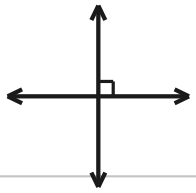
(parallel) perpendicular neither

13) 

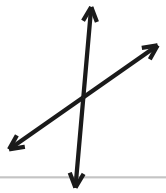
parallel perpendicular **(neither)**

14) 

(parallel) perpendicular neither

15) 

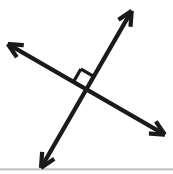
parallel **(perpendicular)** neither

16) 

parallel perpendicular **(neither)**

17) 

parallel perpendicular **(neither)**

18) 

parallel **(perpendicular)** neither

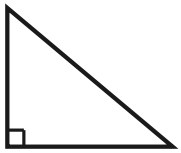
Lines and Shapes: Practice (continued)

Part 2 Parallel, perpendicular or neither? (continued)

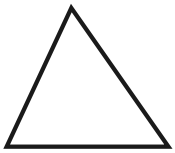
19) 

(parallel) perpendicular neither

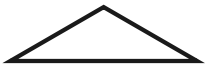
Part 3 Right, acute or obtuse triangle? Look at the biggest corner.

20) 

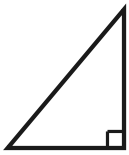
(right) acute obtuse

21) 

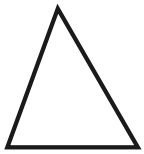
right (acute) obtuse

22) 

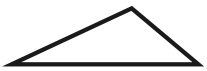
right acute (obtuse)

23) 

(right) acute obtuse

24) 

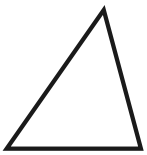
right (acute) obtuse

25) 

right acute (obtuse)

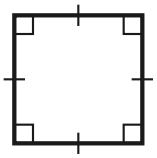
26) 

(right) acute obtuse

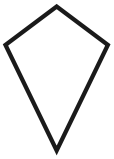
27) 

right (acute) obtuse

Part 4 Lines of symmetry How many ways can you fold it so the halves match?

28) 

lines of symmetry: [4]

29) 

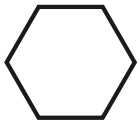
lines of symmetry: [1]

Lines and Shapes: Practice (continued)

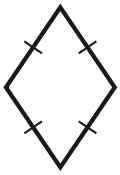
Part 4 Lines of symmetry (continued)

30) 

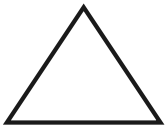
lines of symmetry: 0

31) 

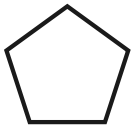
lines of symmetry: 6

32) 

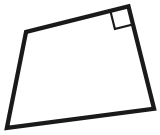
lines of symmetry: 2

33) 

lines of symmetry: 1

34) 

lines of symmetry: 5

35) 

lines of symmetry: 0

Part 5 True or not? Circle yes or no.

36) **A square has 4 right angles.** (yes) no

37) **Parallel lines cross.** yes (no)

38) **A line has two endpoints.** yes (no)

39) **A ray has one endpoint.** (yes) no

40) **A rectangle has 2 pairs of parallel sides.** (yes) no

Lines and Shapes: Practice (continued)

Part 5 True or not? (continued)

41) **A right triangle has 2 right angles.** yes (no)

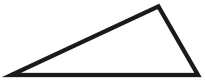
42) **Perpendicular lines make square corners.** (yes) no

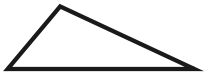
43) **A circle has no lines of symmetry.** yes (no)

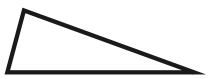
Part 6 What kind of triangle?

44)  right acute (obtuse)

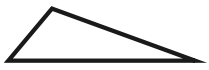
45)  right acute (obtuse)

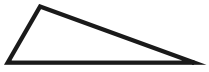
46)  right acute (obtuse)

47)  right acute (obtuse)

48)  right (acute) obtuse

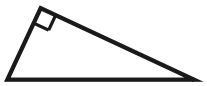
49)  right (acute) obtuse

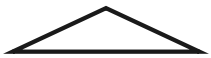
50)  right acute (obtuse)

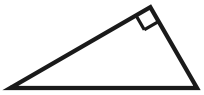
51)  right acute (obtuse)

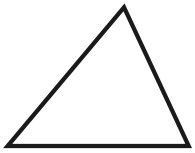
Lines and Shapes: Practice (continued)

Part 6 What kind of triangle? (continued)

52)  (right) acute obtuse

53)  right acute (obtuse)

54)  (right) acute obtuse

55)  right (acute) obtuse

Part 7 Remember? Problems from earlier packets.

56) $40 + \boxed{50} = 90$ degrees

57) $2\frac{1}{8} + \frac{5}{8} = \boxed{2}\frac{\boxed{6}}{8}$

58) area: $21 \times 4 = \boxed{84}$ sq m

59) 7 ft = $\boxed{84}$ in

60) $0.19 = \frac{\boxed{19}}{100}$

61) $63 \times 22 = \boxed{1{,}386}$

Lines and Shapes: Practice (continued)

Part 8 Word problems Draw a picture.

62) A window has 4 square corners. How many right angles does it have?

Number sentence **4 square corners = 4 right angles**

Answer **4** right angles

Draw or show your work

63) Two train tracks never meet. Are they parallel or perpendicular?

Number sentence **never meet = parallel**

Answer **parallel**

Draw or show your work

64) A stop sign has 8 equal sides. How many lines of symmetry does it have?

Number sentence **8 equal sides = 8 lines**

Answer **8** lines

Draw or show your work

65) Challenge: A triangle has corners of 35 and 45 degrees. Is it acute, right or obtuse?

Number sentence **180 − 35 − 45 = 100**

Answer **obtuse**

Draw or show your work

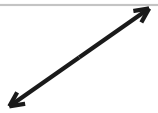
Check-In: Lines and Shapes

Show what you know. No help this time.

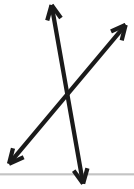
Mastery: 7 of 8 (80%)

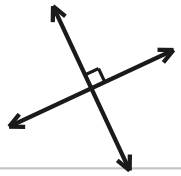
Part 1 Segment, ray or line?

1)  segment (ray) line

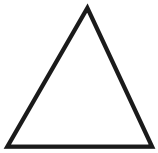
2)  segment ray (line)

Part 2 Parallel, perpendicular or neither?

3)  parallel perpendicular (neither)

4)  parallel (perpendicular) neither

Part 3 Right, acute or obtuse triangle?

5)  right (acute) obtuse

6)  right acute (obtuse)

Part 4 True or not? Circle yes or no.

7) **An obtuse angle is more than 90 degrees.** (yes) no

8) **A rhombus always has a right angle.** yes (no)